

# OSHAN MUDANNAYAKE

Researcher  
Computer Science

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Website [ivantha.com](http://ivantha.com)  
Other  [Google Scholar](#) |  [GitHub](#) |  [Kaggle](#) |  [Medium](#) |  [LinkedIn](#)

## RESEARCH INTERESTS

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Reinforcement Learning | Machine Learning | Artificial Intelligence  
Computer Vision

## EDUCATION

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**[Reading] University of Moratuwa** *Jan. 2023 - 2026*  
M.Sc in Data Science & AI  
Current GPA: 4.02/4.2

**University of Colombo School of Computing** *Feb. 2016 - Feb. 2020*  
B.Sc (Hons) in Computer Science  
First Class Honors with a GPA of 3.61/4.0 (Top 5%)  
Ranked 6<sup>th</sup> over 156 CS students  
Thesis: “*Summarization of Large Scale Streaming Graphs*” (Grade A)  
Activities: Chair - IEEE Student Branch, Core Team - Mozilla Club, Member - CompSoc, Instructor - Exploration Club, Batch Representative - IUD

**Maliyadeva College** *Jan. 2001 - Dec. 2014*  
G.C.E. Advance Level Examination - Physical Science Stream  
Activities: President - Robotics Club, Lead - Hardware and Networking Group

## PUBLICATIONS

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- [1] **[Journal]** O. Mudannayake, A. Indika, U. Jayasinghe, G. M. Lee, and J. Alawatugoda, “On Privacy-Preserved Machine Learning Using Secure Multi-Party Computing: Techniques and Trends,” in *Computers, Materials & Continua (CMC)*, vol. 85, no. 2, pp. 2527-2578, 2025, [doi: [10.32604/cmc.2025.068875](https://doi.org/10.32604/cmc.2025.068875)].
- [2] **[Conference]** S. Singh, ..., O. Mudannayake, and others, “Aya dataset: An open-access collection for multi-lingual instruction tuning,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. [doi: [10.18653/v1/2024.acl-long.620](https://doi.org/10.18653/v1/2024.acl-long.620)]
- [3] **[Journal]** H. Samaranayake, O. Mudannayake, D. Perera, P. Kumarasinghe, C. Suduwella, K. De Zoysa, and P. Wimalaratne, “Detecting Water In Visual Image Streams Captured From Unmanned Aerial Vehicles”, in *Journal of Visual Communication and Image Representation (JVCI)*. [doi: [10.1016/j.jvcir.2023.103933](https://doi.org/10.1016/j.jvcir.2023.103933)].
- [4] **[Journal]** O. Mudannayake\*, D. Rathnayake\*, J. D. Herath, D. K. Fernando, and M. Fernando, “Exploring Machine Learning and Deep Learning Approaches for Multi-Step Forecasting in Municipal Solid Waste Generation,” in *IEEE Access*, vol. 10, pp. 122570-122585, 2022, [doi: [10.1109/ACCESS.2022.3221941](https://doi.org/10.1109/ACCESS.2022.3221941)].
- [5] **[Conference]** O. Mudannayake and N. Ranasinghe, “kMatrix: A Space Efficient Streaming Graph Summarization Technique,” 2021 10th International Conference on Information and Automation for Sustainability (ICIAfS), 2021, pp. 161-166, [doi: [10.1109/ICIAfS52090.2021.9606137](https://doi.org/10.1109/ICIAfS52090.2021.9606137)].

\* Equal contribution

## PREPRINTS

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- [1] Sahan Dissanayaka\*, O. Mudannayake\*, Thilina Halloluwa, and Chameera De Silva, “ImageLab: Simplifying Image Processing Exploration for Novices and Experts Alike”.
- [2] O. Mudannayake, K. Senarathne, S. Farook, L. De Mel, and C. Jayanaka, “Leveraging Spatial and Temporal Data for Enhanced Wildfire Spread Prediction”.

## PROFESSIONAL EXPERIENCE

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|                                                                                                            |                                            |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Lead Machine Learning Engineer (Full-time)</b><br><i>Skillful.ly</i>                                    | <i>Sep. 2023 - Present</i><br>USA - Remote |
| <b>Senior Data Science Engineer (Full-time)</b><br><i>WSO2</i>                                             | <i>Nov. 2021 - Aug. 2023</i><br>Sri Lanka  |
| <b>Data Science Engineer (Full-time)</b><br><i>WSO2</i>                                                    | <i>Apr. 2021 - Oct. 2021</i><br>Sri Lanka  |
| <b>Software Engineer - IAM Team (Full-time)</b><br><i>WSO2</i>                                             | <i>Mar. 2020 - Mar. 2021</i><br>Sri Lanka  |
| <b>Student Developer</b><br><i>CERN-HSF - Google Summer of Code</i>                                        | <i>May. 2018 - Aug. 2018</i><br>Remote     |
| <b>Software Developer (Contract)</b><br><i>Language Matters</i><br>· Supervised by Dr. Leonie Solomons     | <i>Jan. 2018 - Mar. 2020</i><br>Sri Lanka  |
| <b>Software Developer (Contract)</b><br><i>Dreamteam IT Solutions</i><br>· Supervised by Dr. Gihan Kuruppu | <i>Feb. 2016 - Jan. 2017</i><br>Sri Lanka  |
| <b>Course Instructor (Contract)</b><br><i>Dreamteam IT Solutions</i>                                       | <i>Jan. 2016 - Jun. 2016</i><br>Sri Lanka  |

## RESEARCH EXPERIENCE

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Community Researcher (Part-time)</b><br><i>Cohere For AI</i><br>· Advised by Sara Hooker                                                               | <i>Mar. 2023 - Apr. 2024</i><br>Sri Lanka |
| <b>Research Assistant (Part-time)</b><br><i>University of Colombo School of Computing</i><br>· Advised by Dr. Noel Fernando                               | <i>May. 2021 - Dec. 2022</i><br>Sri Lanka |
| <b>Researcher (Part-time)</b><br><i>Sustainable Computing Research (SCoRe) Lab</i><br>· Advised by Dr. Mohamed Nabeel                                     | <i>Aug. 2020 - Dec. 2021</i><br>Sri Lanka |
| <b>Research Intern (Full-time)</b><br><i>University of Colombo School of Computing - Distributed Computing Group</i><br>· Advised by Dr. Nalin Ranasinghe | <i>Jul. 2018 - Jan. 2019</i><br>Sri Lanka |

## SCHOLARSHIPS & GRANTS

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- [1] AutoML Fall School Scholarship, 2021
- [2] Dataquest & AI Inclusive Scholarship, 2021
- [3] Tokyo Data Science Full Scholarship, 2020
- [4] Commonwealth of Learning Coursera Scholarship by Commonwealth of Learning, Canada, 2020
- [5] HackZurich Travel grant, 2019
- [6] AI Summer School Singapore Scholarship<sup>1</sup>, 2019
- [7] ownCloud Conference Travel grant by CERN and ownCloud, 2018
- [8] Hack the North Travel grant, 2018

## HONOURS & AWARDS

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- [1] Merit Award at Datathon (SLIIT Codefest) by Sri Lanka Institute of Information Technology, 2019
- [2] \*IEEE Regional Exemplary Student Branch 2020 under Region 10 and Sri Lanka Section by IEEE, 2019
- [3] \*Bronze category winner of IEEE Darrel Chong Student Activity Award 2020 by IEEE, 2019

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<sup>1</sup>I did not accept this scholarship as I could not attend the event due to university examinations.

- [4] \*Best Student Branch Activity Award for GameRush at IEEE SLSYWC 2019 by IEEE, 2019
  - [5] 1<sup>st</sup> place at Codezilla by Mozilla Campus Club of Uva Wellassa University, 2018
  - [6] 1<sup>st</sup> place at LetMeHack by Sabaragamuwa University, 2018
  - [7] Award of Appreciation at University of Colombo School of Computing on recognizing the outstanding achievements in hackathons and coding competitions., 2017
  - [8] †Best Industry Collaborative Award at IEEE SLSYWC 2017 by IEEE, 2017
  - [9] ‡Best Student Branch Activity Award at IEEE SLSYWC 2017 by IEEE, 2017
  - [10] 2<sup>nd</sup> place at Presistent Hackathon by Persistent Systems Lanka, 2017
  - [11] 2<sup>nd</sup> place at HACKiNTRA by University of Jaffna, 2017
  - [12] Finalist at CodeFest Designathon by Sri Lanka Institute of Information Technology, 2017
  - [13] Finalist at SS12 Asia by IEEE, 2017
  - [14] 3<sup>rd</sup> place at Dialog Game Dev Hackathon by Dialog Axiata, 2017
  - [15] 1<sup>st</sup> place at Mozilla Web Maker Party by Mozilla Club of University of Colombo School of Computing, 2016
  - [16] 2<sup>nd</sup> place at CodeFest Tertiary Hackathon by Sri Lanka Institute of Information Technology, 2016
  - [17] 1<sup>st</sup> place at IEEE preXtreme 1.0 by IEEE Student Branch of University of Colombo School of Computing, 2016
  - [18] 6<sup>th</sup> place at ACES Coders by University of Peradeniya, 2016
  - [19] 5<sup>th</sup> place at Code Expertz by University of Moratuwa, 2016
- \* President of the IEEE Student Branch of UCSC  
† Finance Team Lead of IEEE TAP project  
‡ Program Team Lead of CloudHack 17 hackathon

## POSTER PRESENTATIONS

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- [1] Machine Learning over Encrypted Data, Sri Lanka Association for the Advancement of Science (SLAAS) Exhibition, Sri Lanka, Dec. 2018

## ORAL PRESENTATIONS

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- [1] kMatrix: A Space Efficient Streaming Graph Summarization Technique, IEEE ICIAfS, Sri Lanka, Jul. 2021

## INVITED TALKS

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- [1] Invited speaker, Career guidance for undergraduates by UCSC ACM Student Chapter at University of Colombo School of Computing, Sri Lanka, 2021
- [2] A new CORS Valve for the WSO2 Identity Server by IAM Community Call by WSO2 at WSO2, Sri Lanka, 2020
- [3] Invited speaker, Rust 101 by Mozilla Club of UCSC at University of Colombo School of Computing, Sri Lanka, 2019
- [4] Invited speaker, Lightning talk on Phoenix by ownCloud Conference 2018 at ownCloud GmbH, Germany, 2018

## WORKSHOPS ATTENDED

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- [1] International Conference on Learning Representations (ICLR), May. 2023
- [2] Online Asian Machine Learning School at 13th Asian Conference on Machine Learning, Nov. 2021
- [3] AutoML Fall School, Nov. 2021
- [4] ACM Europe Summer School on HPC Computer Architectures for AI and Dedicated Applications by Barcelona Supercomputing Center and Universitat Politècnica de Catalunya, Aug. 2021
- [5] AI Summer School Singapore 2021, Jul. 2021
- [6] Qiskit Global Summer School 2021: Quantum Machine Learning, Jul. 2021
- [7] AI Summer School Singapore 2020, Aug. 2020

## SELECTED PROJECTS

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### Realtime Property Evaluation of Large Streaming Graphs

*Graphs, Graph Summarization, Python*

Developed a system for summarizing graph streams, addressing a significant problem in the field of large-scale graph analysis and demonstrating the potential of realtime property evaluation for large-scale graphs.

### TessTA

*ML, OCR, React, Java, Spring Boot, Hibernate, Tesseract, AWS*

Created an OCR solution for the Sinhala language utilizing the Tesseract OCR Engine.

## Machine Learning over Encrypted Data

*ML, Encryption, Tensorflow, Python, Google Cloud Platform*

Worked on developing a layer over Tensorflow that enabled machine learning on encrypted data across a distributed network of machines.

## Intelligent Traffic Light System

*Tensorflow, Python, Angular, NodeJS, Raspberry Pi, Arduino, Ionic Framework, GCP*

Automated a traditional time interval based traffic light system to improve traffic management by removing the need for traffic officers and replacing the round robin light schedule with an adaptive system.

## Bra Elasticity Analyzing System

*Image processing, OpenCV, Java, Swing*

Developed a method for comparing the elasticity of sports bras using image processing.

## Yarn Detection System

*Image processing, OpenCV, Python, Raspberry Pi*

Developed a solution to detect the knots in yarn threads used in glove weaving machines, resulting in significant savings of hundreds of needles wasted per day and reduced hours of manpower.

## Glove Defect Identification System

*Image processing, OpenCV, Python*

Utilized image processing to identify common defects in gloves as they were being conveyed on a conveyor belt, enabling the automatic rejection of defective gloves without human intervention, and automating the entire pipeline.

## Smart Glove

*Java, JavaFX, FSR 400, Arduino*

Designed a glove prototype that measures the pressure distribution and acceleration of the hand to derive optimal patterns of movement for cricket players, taking into account the hand pressure.

## CERTIFICATIONS

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- [1] Advanced Machine Learning on Google Cloud Specialization by Coursera, 2021
- [2] Data Engineering, Big Data, and Machine Learning on GCP Specialization by Coursera, 2021
- [3] Machine Learning on Google Cloud Specialization by Coursera, 2021
- [4] Deep Learning Specialization by Coursera, 2020
- [5] Natural Language Processing Specialization by Coursera, 2020
- [6] DeepLearning.AI TensorFlow Developer Specialization by Coursera, 2020
- [7] Generative Adversarial Networks (GANs) Specialization by Coursera, 2020
- [8] Qwicklabs quests on ML and Data Science -by Qwicklabs, 2020
- [9] Oracle Certified Professional Java SE 6 Programmer by Oracle, 2015
- [10] Robotics & Automation by Dreamteam IT Solutions, 2015
- [11] Professional Certified Java Developer by Dreamteam IT Solutions, 2015
- [12] Diploma in Software Engineering by Esoft, 2013
- [13] Diploma in PC Hardware and Network Administration by Esoft, 2010

## MENTORING

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- [1] Mentor at SCoRe Lab in Google Summer of Code, 2019, 2020, 2021, 2022
- [2] Mentor at ReidCon by UCSC, 2021
- [3] Mentor at SCoRe Lab in Google Code-In, 2016, 2017, 2018
- [4] Mentor at 2nd Year Group Project Mentoring by IEEE Student Branch of UCSC, 2018
- [5] Question Panel Member - preXtreme 2.0 - IEEE Student Branch of UCSC, 2017
- [6] Mentor at International School Software Competition (ISSC) by Computer Society of Sri Lanka, 2017
- [7] Mentor at Help to Code by Information and Communication Technology Agency of Sri Lanka, 2017
- [8] Mentor at National School Software Competition (NSSC) by Computer Society of Sri Lanka, 2016

## OPEN SOURCE CONTRIBUTIONS

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### WSO2 Identity Server

*Java, OSGi, REST, Tomcat Valves*

Designed and shipped the tenant-aware CORS architecture for Identity Server 5.11 end-to-end – OSGi management

service, DAO layer, REST APIs (tenant, application, and origin scopes), servlet valve, integration tests, and the public migration guide — landing changes across [carbon-identity-framework](#), [identity-api-server](#), [product-is](#), and [docs-is](#).

### SCoRe Lab (Sustainable Computing Research Group)

*Python, Flask, React, Redux, Elasticsearch, TensorFlow*

Contributed as a Google Summer of Code student developer and returning mentor across multiple years — built the authentication and news-crawler stack for [fact-bounty](#), bootstrapped the drag-and-drop ML client for [TensorMap](#), and shepherded code contributions into [DroneSym](#) and [ImageLab](#).

### C2SI — Ceylon Computer Science Institute

*GitHub Actions, CI/CD, Python*

Maintain and modernise the CI pipelines for the successor organisation of SCoRe Lab projects — hardened fork-PR linting, slash-command automation triggers, and release workflows across [ImageLab](#) and [TensorMap](#).

### Cohere Labs Community — Instruct Multilingual

*Next.js, Nest.js, PostgreSQL, Docker, Google Cloud Platform*

Built and deployed the open-source web platform used to collect multilingual instruction-tuning data from a global contributor community — including Google and Discord SSO, the full task-annotation workflow, analytics, RTL language support, and the Cloud Build deployment pipeline.

## VOLUNTEERING & COMMITTEE POSITIONS

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- [1] Volunteer at ICLR, 2023
- [2] Judge panel member (Area: ML) at IntelliHack by IEEE CS Chapter of UCSC, 2023
- [3] Judge panel member at IEEE YESIST12 2021 Junior Einstein Track by IEEE, 2021
- [4] Judge panel member (Area: ML) at MadHack by IEEE Student Branch of UCSC, 2021
- [5] Core Team Member at Google Developer Group (GDG) Sri Lanka, 2019
- [6] President of IEEE Student Branch of UCSC, 2019
- [7] Team Lead at Mozilla activate Sri Lanka - Rust Team, 2019
- [8] Web Master of IEEE Sri Lanka Section - Student activity Committee (SAC), 2018
- [9] Secretary of IEEE Student branch of UCSC, 2018
- [10] Operational Committee Member at Mozilla Activate Sri Lanka, 2018
- [11] Android Developer at Global Geek Sustainable Learning Kingdom (GGSLK) Sri Lanka, 2018
- [12] Operational Committee Member at Program Team - IEEE Student/YP/WIE Congress, 2017
- [13] Finance Team Lead at Technological Awareness Project by IEEE Sri Lanka Section, 2017
- [14] Program Team Lead at CloudHack 17 by IEEE Student Branch of UCSC, 2017
- [15] Core Team Member at Mozilla Club UCSC, 2017
- [16] Coordinator at IEEE Student Branch of UCSC, 2017
- [17] Operational Committee Member at Finance Team - Innover (IEEE IAS IOT Workshop), 2016
- [18] Operational Committee Member at Program Team - IEEE Day UCSC, 2016
- [19] Instructor at UCSC Exploration Club, 2016
- [20] President of Robotics Club at Maliyadeva College, 2012, 2013